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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Regular / Supplementary Examinations Feb /Mar -2024

Computer Science and Engineering**PARALLEL ARCHITECTURE AND DISTRIBUTED PROGRAMMING****Time: 03 Hours****Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	_____ exploits parallelism among largely decoupled tasks specified by the programmer or the operating system.	01									
	1.2	Define dynamic energy in CMOS chips.	02									
	1.3	Define the term 'basic block' with respect to instructions level parallelism.	01									
	1.4	Differentiate between total store ordering from partial store order.	02									
	1.5	Assuming that words x1 and x2 are in the same cache block that is in shared state in the caches of both P1 and P2, identify the following events as a true sharing miss, a false sharing miss , or a hit.										
		<table><tr><td><i>Time</i></td><td><i>P1</i></td><td><i>P2</i></td></tr><tr><td>1</td><td><i>Write</i></td><td><i>x1</i></td></tr><tr><td>2</td><td><i>Read</i></td><td><i>x2</i></td></tr></table>	<i>Time</i>	<i>P1</i>	<i>P2</i>	1	<i>Write</i>	<i>x1</i>	2	<i>Read</i>	<i>x2</i>	02
	<i>Time</i>	<i>P1</i>	<i>P2</i>									
	1	<i>Write</i>	<i>x1</i>									
	2	<i>Read</i>	<i>x2</i>									
	1.6	Mention any two primary components of the instruction set architecture of <i>VMIPS</i> .	02									
	1.7	Define Convoy, identify the number of convoys required for the following code sequence.										
		<table><tr><td><i>LV</i></td><td><i>V1, Rx</i></td></tr><tr><td><i>MULVS.D</i></td><td><i>V2, V1, F0</i></td></tr><tr><td><i>LV</i></td><td><i>V3, Ry</i></td></tr><tr><td><i>ADDVV.D</i></td><td><i>V4, V2, V3</i></td></tr><tr><td><i>SV</i></td><td><i>V4, Ry</i></td></tr></table>	<i>LV</i>	<i>V1, Rx</i>	<i>MULVS.D</i>	<i>V2, V1, F0</i>	<i>LV</i>	<i>V3, Ry</i>	<i>ADDVV.D</i>	<i>V4, V2, V3</i>	<i>SV</i>	<i>V4, Ry</i>
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1.8	Name two additional datatype that MPI allows which are not part of the C language.	02										
1.9	The number and size of tasks into which a problem is decomposed determines the _____ of the decomposition.	01										
1.10	State Owner-computes rule.	01										
1.11	In OpenCL, the kernel is launched with a two-dimensional _____ containing all work items.	02										
1.12	In the OpenCL, the kernel calls the <i>API</i> functions _____ & _____ to fetch the total number of work items in the x- and y-dimensions of all work items.	02										

PART-B

2	a	Discuss Andahl's law along with the factors that are needed to find the speedup from enhancement.	06
	b	Briefly describe the approaches employed to achieve $CPI < 1$ while implementing instruction level parallelism (<i>ILP</i>)?	06
	c	How many bits are in the (0,2) branch predictor with 4K entries? How many entries are in a (2,2) predictor with the same number of bits?	04
3	a	Discuss and differentiate centralized shared memory from distributed memory multiprocessors with a suitable basic structure for the same.	08
	b	Discuss briefly the hardware primitives for implementing synchronization in multiprocessors. Give the code sequence that implements an atomic exchange by employing a pair of these special instruction (an atomic exchange on the memory location).	08
OR			
4	a	Explain snooping cache coherence for write invalidate protocol with a neat state transition diagram.	10
	b	Discuss briefly two most common extensions to the basic coherence protocol.	06
5	a	Draw the basic structure of vendor architecture for <i>VMIPS</i> and explain its components.	08
	b	Let X and Y are vectors, initially resident in memory and 'a' is a scalar, the following instruction forms the inner loop of the Linpack benchmark. Assume that the starting addresses of X and Y are in Rx and Ry, respectively. Show the code for <i>MIPS</i> and <i>VMIPS</i> for the given loop.	08
$Y = a * X + Y$ OR			
6	a	Describe gather-scatter operations using index vectors for handling sparse matrices in vector architectures.	05
	b	Write <i>CUDA</i> code snippet for the <i>DAXPY</i> loop given below, to launch n threads, one per vector element, with 256 <i>CUDA</i> Threads per thread block in a multithreaded <i>SIMD</i> Processor.	04
	c	With a neat diagram explain the process of scheduling threads of <i>SIMD</i> instructions.	07
7	a	Differentiate between blocking and non-blocking message passing operations.	10
	b	Discuss the characteristics of inter-task interactions highlighting the salient aspects of each.	06
8	a	Describe OpenCL model for device management with a suitable diagram and simple host program to create a context for the device.	08
	b	Write a OpenCL kernel function for vector addition and discuss OpenCL parallel execution model with a neat diagram for the same.	08